

TOBIN CARLTON VAN BUIZEN

Mathematics Honours and Statistics, Australian National University.

Research interests: algebraic topology

[Accelerated entry](#): commenced university at 15; completed Honours year at 20.

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EDUCATION	The Australian National University	Canberra, Australia
	<i>Bachelor of Mathematical Sciences (Honours)</i>	2021 – 2025
	• First Class Honours • Supervisor: Dr. Vigleik Angeltveit. • Research area: Equivariant Stable Homotopy Theory • Descriptive course list (upper-division/graduate): Available here.	
	The Australian National University	Canberra, Australia
	<i>Bachelor of Statistics</i>	2021 – 2024
	• Descriptive course list (upper-division/graduate): Available here.	
	Australian Mathematical Sciences Institute	Canberra, Australia
	<i>Summer School 2024</i>	2023 – 2024
RESEARCH PROJECTS	• Descriptive course list: Available here.	
	Honours Thesis: Twisted Topological Hochschild Homology	
	<i>Australian National University</i>	2025.01 – 2025.11
	• Supervisor: Dr. Vigleik Angeltveit. • Developed a twisted model of p-cyclotomic spectra and twisted TC for C_n-ring spectra. • Provided a Nikolaus-Scholze style theorem for computing twisted TC. • Provided an identification between norms and twisted cyclic bar constructions.	
	Odd Primary Hopf Invariant One	
	<i>Australian National University</i>	2025.02 – 2025.06
	• Surveyed odd-primary refinements and constraints on Hopf invariant one.	
	Lévy Integration	
	<i>Australian National University</i>	2024.08 – 2024.11
	• Constructed Lévy-Itô integration for general Lévy processes and compare with Itô integration. • Provided worked examples for compound Poisson and variance-gamma processes.	
	Applications of Differential Forms to Homotopy Theory	
	<i>Australian National University</i>	2024.08 – 2024.11
	• Surveyed rational homotopy theory and differential graded algebras.	
	Kervaire Invariant One: C_2-Equivariance and KR-Theory	
	<i>Australian National University</i>	2024.02 – 2024.06
	• Expository project connecting C_2-equivariant techniques to the Hill-Hopkins-Ravenel approach.	
	UMD Spaces	
	<i>Australian National University</i>	2023.08 – 2023.11
	• Expository project on spaces with unconditional martingale differences, and applications to stochastic calculus.	

TEACHING	Demonstrator The Australian National University 2023.07 – Present	
	• MATH1005: Discrete Mathematical Models (2023.07 – 2024.11) - ran workshops, marked assignments, marked exam and wrote solutions. • MATH1115: Advanced Mathematics and Applications 1 (2026.02 – 2026.06) - running workshops, marking assignments and marking exams. • MATH2222: Introduction to Mathematical Thinking (2025.02 – 2025.06) - ran workshops, marked assignments and marked exam. • MATH4204: Algebraic Topology (Honours) (2025.07 – Present) - marked assignments. • MSI Drop-In Help Centre MSI Drop-In Help Centre (2026.08 – Present) - answering questions from students in all MATH courses at ANU, primarily first and second year.	
	TALKS	THH with a “Twist” - Honours Conference, ANU 2025.11
		What is... <i>G</i> -stuff? - What is...? Seminar, ANU 2025.08
		Odd Primary Hopf Invariant One - VBKT Seminar, ANU 2025.06
CONFERENCE PARTICIPATION		The Scariest Maths Topics (in 8 minutes) - Maths in the Pub (ANU MSI) 2025.05
		Lévy Integration - Stochastic Analysis & Financial Applications 2024.11
		C_2 -Equivariance and KR -Theory - SHT Seminar, ANU 2024.06
		Workshop on Interactions between Group Theory and Homotopy Theory - Fields Institute 2025.07
		Graduate School on Polyhedral Products (Steven Amelotte) - Fields Institute 2025.07
SERVICE		Graduate School on Chromatic Homotopy Theory (Agnès Beaudry) - Fields Institute 2025.07
		Co-organiser, “What is...?” student seminar, ANU (2025)
WORK EXPERIENCE	Evolve Capabilities	Canberra, Australia
	<i>Data Analyst</i>	2026.01 – Present
	Australian Public Service	Canberra, Australia
SKILLS	<i>Configuration Management & Reporting Service Delivery Manager</i>	2023.02 – 2026.01
	Programming: Python, R, MATLAB, HTML, CSS; SQL, Excel, PowerBI, PowerQuery	
	Typesetting: L ^A T _E X (BibTeX, TikZ, TikZ-cd)	
AWARDS	Languages: English (native); some Chinese	
	• Caltex Best All Rounder Award - Presented in recognition of excellence in academic, leadership, sporting and community service activities. • Erindale Sports Academy Academic Excellence Award - Presented by Erindale College (2020). • Australian Representatives Honour Roll - Recognised by Erindale College for representing Australia in baseball (2019). • Member of MENSA Australia.	